Paper Review

MS-HuBERT

1. Paper Details

- **Title of the Paper:** MS-HuBERT: Mitigating Pre-training and Inference Mismatch in Masked Language Modelling methods for learning Speech Representations
- **Authors:** Hemant Yadav, Sunayana Sitaram, Rajiv Ratn Shah
- **Venue:** Interspeech-2024
- **Link:** https://www.isca-archive.org/interspeech_2024/yadav24b_interspeech.html

2. Summary of the Paper

The paper titled as above, proposes a novel advancement over existing HuBERT for generating better and refined speech representation. It is a self supervised learning method which addresses the issue of mismatch between pre-training and inference phase. It proposes two techniques which work over the same architecture as HuBERT, which are Swap method and Multicluster Masked Prediction Loss (MPL). The authors evaluated the idea with 4 different approaches. The major ones are through using supervised finetuning / inference and using MS-HuBERT as feature extractor. MS-HuBERT performed significantly well over HuBERT and performed close to data2vec on the ASR Librispeech benchmark especially in high-resource settings.

3. Main Figure from the Paper

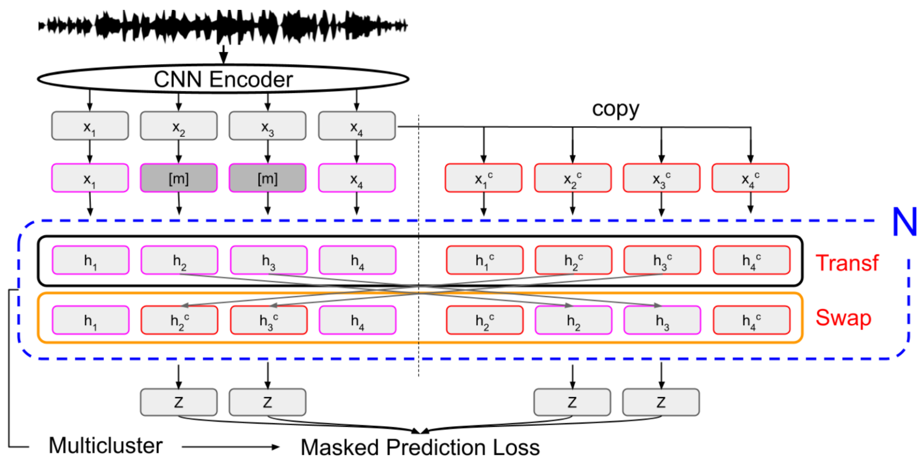

Figure 1: *MS-HuBERT architecture: raw audio is encoded by CNN, split into masked/unmasked views, passed through a Swap-enabled transformer encoder, and trained with Multicluster Masked Prediction Loss.*

4. Methodology Overview

MS-HuBERT improves upon existing HuBERT framework having 2 encoders. The first encoder is CNN and second encoder is transformer. Because during inference, masking is not done, the mismatch between pre-training and inference phase occurs in HuBERT. To rectify this, two core improvements are done in MS-HuBERT. Firstly, two views of output of first encoder is created using unmasked and masked view(50% token masked). This batch of 2 is passsed to second encoder. In Swap method, in second encoder, the output of each view is swapped and given to next layer. In Multicluster MPL, loss is computed at multiple layers of transformer encoder using 6 pseudo labels generated from first iteration HuBERT thus increasing model capacity utilization. In Pre-training, MS-HuBERT base has 6 classification heads and in supervised finetuning and inference, Wave2Vec2 strategy of reducing Connectionist Temporal Classification(CTC) loss is used. Thus, the swap method enhanced the context exposure in analogy to idea of genetic mutation and improves generalization at inference. And the Multicluster Masked Prediction Loss (MPL), using different resolution clusters(1000,500,250,125,50,25) as pseudo labels enhances level of granuality of supervision at each layer.

5. Experiments & Results

- ASR Librispeech benchmark dataset is used for both pre-training and finetuning stage. For pre-training raw audios from the combined training split i.e. total 960 hours audio is used. For supervised fine-tuning, three sets of Libri-Light: 1 hour, 10 hour, 100 hour and the full Librispeech 960 hours of dataset is used.
- Word Error Rate (WER) is the main evaluation metric.

- In Supervised Fine-tuning and Inference stage, MS-HuBERT is compared against HuBERT and WavLM.
- When evaluating as feature extractor, MS-HuBERT is compared against HuBERT, WavLM, wav2vec 2.0, and data2vec. The author used two experiments i.e. SUPERB benchmark(ASR and phoneme-recognition (PR) tasks) and canonical correlation analysis (CCA) similarity with word labels.
- The results shows significant advancement over HuBERT and comparable performance to data2vec, at larger data scales.
- Also, ablation studies confirmed that both Swap and Multicluster components contribute to gains, with Swap method contributing immensely.

6. Strengths of the Paper

- Two view(masked and unmasked) approach and swapping after each layer of transformer encoder ensuring context capturing
- Multicluster MPL introducing hierarchical supervision and improving model capacity usage.
- Comparable performance to data2vec in ASR/LR task at large scale data
- Comparable performance to WavLM base + at low scale data as shown in 3rd iteration models
- For PR task weighted average of outputs from all layers can be taken and for ASR task specific layer output can be taken thus reducing transformer encoder computation.

7. Weaknesses of the Paper

- Swap and Multicluster MPL increased computational cost during pre-training but same computation during inference as HuBERT.
- Too much dependency on pseudo-label quality from initial HuBERT iterations.
- Lagging in performance in low resource setting to data2vec.
- Iterative process of label refinement.
- Stepsize selection of MPL layers for loss calculation may affect learning. capacity

8. Minor Questions / Minor Weaknesses

- Hyperparameters tuning not well informed.
- Only one dataset being used for evaluation.
- Affect of stepsizing of MPL layers not analysed.
- Results from pre-training on different dataset and finetuning on different dataset.

- Why labels are used as cluster of 1000-500-250- and so on?
- What is MS-HuBERT + ?

9. Suggestions to the Authors

- The paper can be more beneficial if the authors can discuss the affect of MS-HuBERT representation on other downstream speech tasks.
- Also, an algorithm can be incorporated in the design to decide on the MS-HuBERT(iter 3) pseudo label selection.
- Also initial label selection may have a large affect on the downstream performance. So, authors can suggest ways or empirical results on downstream task due to different selections from HuBERT or self layers.
- The authors can scale the transformer model and comment upon the effects of step size selection during calculation of multicluster MPL.
- To aid reproducibility, the authors can comment upon the estimated dates for releasing pre-trained weights and hyperparameter selections.
- Authors can also mathematically prove the affects of swapping using regularization tricks.

10. Rating and Justification

Rating: Strong Accept
Justification: The paper proposes an indigenous technical method which improves the representation of speech with SOTA performance in the downstream ASR task. They have proved better performance over HuBERT using emperical results and ablation study. Swap and Multicluster MPL are a great step ahead over existing HuBERT , and the methodology is clear and reproducible with minor clarifications.

11. Scorecard

Criterion	Rating (1–5)
Novelty	5
Technical depth	5
Experimental validation	4
Clarity	4
Reproducibility	4
Overall recommendation	5